Table 1.4 - Buffer systems of the body fluids

Sl. No	Body Fluid	Buffer system
1	Blood	Bicarbonate, Protein and Hemoglobin buffer system
2	Interstitial fluid	Bicarbonate buffer system
3	Intracellular fluid	Protein and Phosphate buffer system

Various Buffers of Blood

Blood contains four buffers namely

- Bicarbonate buffer system
- Phosphate buffer system
- Protein buffer system
- Hemoglobin buffer system



The buffering capacity of blood for handling carbon dioxide is estimated to be distributed among various buffer systems as follows: Haemoglobin 62%, Phosphate 22%, Plasma protein 11% and Bicarbonate buffer system 5%.

Bicarbonate buffer system

The bicarbonate buffer system consists of carbonic acid and bicarbonate ions. The pK_a of the bicarbonate buffer system is 6.1. It is the most important buffer system of blood plasma. The carbon dioxide released during fuel metabolism reacts with water by the action of the enzyme carbonic anhydrase to form carbonic acid [H_2CO_3]. Carbonic acid is a weak acid that partially dissociates into bicarbonate ion [HCO_3^-] and H^+ ion.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

As base is added and H^+ removed, carbonic acid dissociates into hydrogen ion and bicarbonate ions, and dissolved CO_2 reacts with water to replenish the carbonic acid levels. When CO_2 levels are increased, it forms more amount of carbonic acid which in turn dissociates into hydrogen ion and bicarbonate ions. Thus bicarbonate buffer functions as buffer system in blood.

Phosphate buffer system

The dihydrogen phosphate [$H_2PO_4^-$] ions and monohydrogen phosphate [HPO_4^{2-}] ions contribute to the phosphate buffer system. The pK_a of a phosphate buffer system is 6.8. Phosphoric acid dissociates into H^+ ions and dihydrogen phosphate [$H_2PO_4^-$] ions with pK_a of 2.15. Dihydrogen phosphate [$H_2PO_4^-$] ion dissociates into H^+ ions and monohydrogen phosphate [HPO_4^{2-}] ions with pK_a of 7.2 whereas monohydrogen phosphate ions dissociates into hydrogen ion and phosphate PO_4^{3-} anions with pK_a of 12.4. From the dissociation constant values, it is clearly understood that phosphate acts as an effective buffer in blood (pH = 7.4).

$$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$$

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